EXPERIMENT 6

Simulate the lifecycle stages for UI design using the RAD model and develop a small interactive interface.

NAME : SUBASH R

ROLL.NO : 2116240701538

Rapid Application Development (RAD) is a highly adaptive project management strategy that prioritizes rapid prototyping and quick feedback loops over the rigid, sequential planning found in traditional "Waterfall" methods. It is the "move fast and break things" approach—except with a structured focus on user satisfaction and functional software.

The Core Philosophy of RAD

Before diving into the phases, it's important to understand what makes RAD different:

- **Prototypes Over Paper:** Instead of a 100-page manual, you build a clickable model.
- **Continuous Feedback:** Users don't wait until the end to see the product; they see it every week.
- **Modular Construction:** Software is built like Lego bricks—reusable and easy to swap.
- **Speed is King:** Ideal for markets where being first matters more than being "perfect" on day one.

Phase 1: Requirements Planning

In this phase, stakeholders and developers sit in a room (or a Zoom call) to define the "What" and the "Why" of the project.

1. Defining the Core Features

The focus is on the essential user journey for an e-commerce platform:

- **Navigation Architecture:** Home, Categories, Product Listings, and Details.

- **The Conversion Funnel:** Cart management, Checkout, and Order Confirmation.
- **User Capabilities:** Searching, filtering, real-time tracking, and account management.

2. The Living Requirements Document

Unlike traditional specs, this list is meant to evolve:

1. Browse categories and filter products.
2. Access deep-dive product information (specs, reviews).
3. Manage a dynamic shopping cart.
4. Secure checkout via shipping/payment gateways.
5. Receive instant digital confirmation.

Phase 2: User Design (Interactive Blueprinting)

This is where the vision becomes visual. Designers and users work together to create the UI/UX architecture.

The Prototyping Toolkit

- **Primary Tool: Axure RP** or **Figma** are used to create high-fidelity, clickable wireframes.
- **Components & Libraries:** Using "Master" components (Headers, Footers, Buttons) to ensure design consistency and speed up the drawing process.

Visualizing the Flow

- **Wireframe Mapping:** Building the skeletons for the Home Page, Product Grids, and Checkout screens.
- **Functional Annotation:** Explaining what happens when a user clicks "Add to Cart"—does a sidebar open? Does a toast notification appear?

Phase 3: Construction (The Iterative Loop)

In RAD, "Construction" doesn't mean writing code in a vacuum. It means building, testing, and refining in a tight loop.

1. Functional Prototyping

The designer links screens to simulate a working app. A user can actually "click" their way from a Product Category to a successful Order Confirmation.

2. Dynamic Interactions

- **States & Modals:** Using dynamic panels to show "Loading" states or "Success" pop-ups without reloading the page.
- **Data Simulation:** Using variables to show the cart total updating in real-time.

3. The Feedback Engine

- **Stakeholder Reviews:** Using Figma's "Present" mode or Axure's Cloud share to get immediate feedback.
- **Rapid Iteration:** If a user finds the checkout flow confusing, the design is changed immediately—not in the next project phase.

Phase 4: Cutover (The Final Hand-off)

This is the transition from a "design prototype" to a "live product."

1. Design-to-Code Export

- **Developer Handoff:** Using tools like **Figma-to-HTML** or **Inspect** mode so developers can grab CSS values, assets, and spacing directly.
- **Documentation:** Providing a "UI Kit" that defines colors, typography, and interaction rules.

2. Launch & Training

- **Stakeholder Walkthroughs:** Final demonstrations to ensure the business logic aligns with the build.
- **Support Documentation:** Using comments and annotations within the design files to act as a manual for the engineering team.

Nozama

Search items

Add Items | Cart | Checkout

Iphone 17 Pro

VIEW ITEM | CART +

Galaxy S26 Ultra

VIEW ITEM | CART +

Pixel 10

VIEW ITEM | CART +

Xiaomi 17 Pro

VIEW ITEM | CART +

Proceed to cart

Home | Cart | Orders | Profile

Nozama

Search items

Add items | Cart | Checkout

$1,399

Galaxy S26 Ultra

REMOVE

Add more items

Proceed to checkout

Home | Cart | Orders | Profile

Nozama

Disabled

Add items | Cart | **Checkout**

Order Summary

Cart Value	**$1399**
Delivery Charges	**$10**
Tax and platform fee	**$11**
Total	**$1420**

Add Delivery Address

Place Order

Home | Cart | **Orders** | Profile